

Indeterminacy as a Scientific Result

A Four-Outcome Framework for Consciousness Attribution

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Abstract

Tests for consciousness are increasingly applied where ordinary verbal report is absent, unreliable, or inapplicable: severe brain injury, infancy, nonhuman animals, neural organoids, and artificial systems. In these cases a negative test is often treated as evidence that consciousness is absent, while conflicting or non-transferable evidence is compressed into a vague category of uncertainty. This paper argues that such binary treatment is scientifically inadequate. Consciousness attribution should use four outcomes: positive, negative, indeterminate, and unlicensed. A result is indeterminate when an applicable and adequately executed assessment yields evidence that does not discriminate between presence and absence. A result is unlicensed when the test's validity conditions are not met, the target is not adequately specified, the system boundary is unresolved, or the test has not been calibrated for the population or substrate. Indeterminate results should additionally carry a cause profile identifying evidential scarcity, surrogate discordance, model disagreement, boundary instability, or transport failure. This framework prevents absence of evidence from being converted into evidence of absence, preserves informative disagreement among indicators, and links each unresolved result to a specific research response. It also separates scientific classification from precautionary action: an indeterminate attribution need not imply inaction, but neither does it scientifically establish consciousness. The proposal applies across competing theories and can be incorporated into both theory-driven and theory-light consciousness tests.

Keywords: consciousness attribution; indeterminacy; C-tests; diagnostic uncertainty; disorders of consciousness; animal consciousness; artificial consciousness; precaution

1. Introduction

The easiest consciousness experiments begin with subjects who can understand instructions, report their experiences, and repeat the task. The hardest cases are precisely those in which these conveniences disappear. A patient may retain awareness while lacking reliable motor output. An infant or nonhuman animal may be unable to understand the reporting protocol. An organoid may have no behavior rich enough to interpret. An artificial system may reproduce human-like reports while differing radically from the systems in which those reports were calibrated. These cases require tests for consciousness, or C-tests, that do not simply presuppose ordinary adult human report (Bayne et al., 2024).

The scientific difficulty is not exhausted by imperfect instruments. Consciousness science lacks a universally accepted theory, a generally applicable gold standard, and a single measurement vocabulary that transfers cleanly across populations and substrates (Bayne et al., 2024; Seth & Bayne, 2022). Behavioral, computational, anatomical, electrophysiological, perturbational, and theory-derived indicators can converge, diverge, or fail to apply. Yet the public and scientific language surrounding these assessments remains strongly binary. A system is described as conscious or nonconscious; a test is said to detect or fail to detect consciousness; an indicator is present or absent. Results that do not fit these categories are often described merely as uncertain, excluded from analysis, or absorbed into the negative class.

This paper argues that indeterminacy should be treated as a scientific result rather than as an embarrassment awaiting deletion. The claim is not that consciousness itself must be metaphysically indeterminate. The claim concerns warranted attribution: given a specified target, system, test, evidence base, and validity envelope, the evidence may fail to license either a positive or a negative classification. That failure can be informative. It may identify inadequate sensitivity, disagreement among credible surrogates, dependence on a contested theory,

instability in the boundary of the candidate system, or failed transfer from the population in which the test was validated.

The proposal has two parts. First, consciousness assessments should return four possible outcomes:

1. **Positive:** the evidence licenses attribution of the target form of consciousness.
2. **Negative:** the evidence licenses denial of the target form of consciousness.
3. **Indeterminate:** the assessment is applicable and adequately executed, but the evidence does not discriminate sufficiently between presence and absence.
4. **Unlicensed:** one or more conditions required to interpret the assessment are not satisfied.

Second, indeterminate and unlicensed outputs should be accompanied by a cause profile. A bare label such as “uncertain” does little scientific work. By contrast, identifying a result as indeterminate because validated measures conflict suggests replication or adjudication; identifying it as unlicensed because the test has not been transported across substrates suggests a calibration program; identifying it as boundary-sensitive suggests interventions on candidate system boundaries.

This structure is intended to be theory-plural rather than theory-neutral in the strong sense. No consciousness test is wholly independent of assumptions about its target and evidential relevance. The framework instead requires those assumptions to be stated and prevents the output of one test from silently acquiring a wider interpretation than its validation permits.

2. Why Binary Classification Fails

2.1 Consciousness tests lack a universal gold standard

Diagnostic science ordinarily evaluates a test against an accepted reference standard. For consciousness, the most secure reference cases are awake, communicative adult humans, but the very need for C-tests arises as systems become less similar to those cases. Even in disorders of consciousness, behavioral examination, task-based neuroimaging, resting-state measures, and perturbational indices have different cognitive demands and different routes to false negatives. Peterson et al. (2016) argue that standard validation is especially difficult where no single measure functions as an adequate gold standard and recommend consilience across partially independent evidence.

The problem is not solved by calling report the gold standard. Report is powerful evidence when it is credible, but failure to report can result from impaired language comprehension, memory, attention, motivation, arousal, or motor output. No-report paradigms were developed partly because neural activity associated with report preparation can be mistaken for activity constitutive of experience (Tsuchiya et al., 2015). In patients with severe brain injury, repeated behavioral assessments improve diagnostic reliability (Wannez et al., 2017), and task-based fMRI or EEG can reveal cognitive motor dissociation in a substantial subset of behaviorally unresponsive patients (Bodien et al., 2024). These findings do not establish that every unresponsive patient is conscious. They establish that behavioral non-response is not, by itself, strong evidence of absence.

2.2 The cost of a false negative is easily hidden

A positive consciousness result usually demands explanation. Investigators ask whether the signal reflects artifact, unconscious processing, task compliance, or overfitting. Negative results often receive less scrutiny. But a

negative output is warranted only if the test had a reasonable opportunity to detect the target. A patient who fails a demanding imagery task may lack the attention, comprehension, memory, or sustained control required by the task while retaining experience. An animal may fail a human-designed paradigm for ecological rather than phenomenological reasons. An AI system may fail a biological marker because the marker is not substrate-general. An organoid may produce signals that resemble a human neural correlate without satisfying the conditions under which that correlate was validated.

The general diagnostic literature distinguishes valid intermediate results from uninterpretable or invalid ones and warns that excluding inconclusive outputs can inflate estimates of accuracy (Shinkins et al., 2013; Simel et al., 1987). Consciousness science needs the same discipline, with an additional complication: errors can arise not only from the test but also from disagreement about the target and the theory connecting evidence to that target.

2.3 Heterogeneous uncertainty is flattened

Consider three cases:

- A well-calibrated perturbational measure returns a value between preregistered rule-in and rule-out thresholds.
- Two independently validated indicators return opposed results.
- A measure validated in adult mammalian brains is applied to a transformer-based artificial system.

All three may be called uncertain, but they are not scientifically equivalent. The first is a valid gray-zone result. The second is unresolved evidential discordance. The third may not be an interpretable consciousness result at all. It is an extrapolation whose validity remains to be established. Pooling these cases conceals what is known, what failed, and what investigation should occur next.

2.4 Disagreement among theories changes the evidential map

Leading theories of consciousness emphasize different properties: integration, global availability, recurrent processing, higher-order representation, predictive organization, or other dynamical and representational features (Seth & Bayne, 2022). Theory-derived indicator approaches can make this disagreement productive by asking whether a system possesses properties associated with several theories (Butlin et al., 2023). But indicator accumulation does not remove model uncertainty. If one theory treats a feature as necessary while another treats it as incidental, the same evidence can support incompatible classifications.

This matters especially beyond familiar biological cases. Similarity to a known conscious system weakens as anatomy, development, embodiment, behavior, and causal organization diverge. A result may therefore be empirically precise while remaining interpretively unstable. Indeterminacy is sometimes located not in the data but in the bridge from data to consciousness.

3. Four Outcomes, Not Two

3.1 The assessment tuple

A consciousness attribution should be indexed to at least five elements:

$$\mathcal{A} = \langle S, C, T, E, \mathcal{V} \rangle,$$

where:

- S is the candidate system at a specified spatial and temporal grain;
- C is the target, such as phenomenal presence, a particular conscious content, wakeful awareness, sentience, or self-consciousness;
- T is the test or assessment battery;
- E is the evidence produced;
- $\mathcal{V}$ is the validity envelope within which the test has an established interpretation.

The validity envelope includes the populations, substrates, task conditions, recording quality, system boundaries, and auxiliary capacities for which the inferential relation between E and C has been justified. A result is not simply “positive for consciousness.” It is positive for a specified target, in a specified system, under an assessment whose scope is explicit.

3.2 A four-outcome decision rule

Let $a(T, S)$ indicate whether the test is applicable to the system and target, and let $q(E)$ indicate whether execution and data quality meet preregistered requirements. Let $L(E; T, C)$ represent the degree of evidential support supplied by the test, with a rule-out threshold τ_- and rule-in threshold τ_+ , where $\tau_- < \tau_+$. Then:

$$D(\mathcal{A}) = \begin{cases} \text{Positive,} & a = 1, q = 1, L \geq \tau_+ \\ \text{Negative,} & a = 1, q = 1, L \leq \tau_- \\ \text{Indeterminate,} & a = 1, q = 1, \tau_- < L < \tau_+ \\ \text{Unlicensed,} & a = 0 \text{ or } q = 0. \end{cases}$$

This representation is schematic. L could be a likelihood ratio, posterior probability, validated score, structured evidence grade, or result of a multi-indicator decision procedure. The framework does not require consensus on Bayesian or frequentist interpretation. It requires explicit thresholds and an explicit distinction between a valid nondiscriminating result and a result that cannot bear the intended interpretation.

Unresolved conflict among measures requires an extension. If credible measures cannot be combined without suppressing material disagreement, the output is indeterminate even when some individual measures cross τ_+ or τ_- . A battery should not be allowed to select whichever surrogate yields the preferred answer after the result is known.

3.3 Positive

A positive result means that the evidence licenses attribution of the specified target under the stated model and validity envelope. It need not imply certainty, theoretical finality, or a complete explanation of experience. A positive command-following result in a behaviorally unresponsive patient can provide strong evidence of preserved awareness without revealing the character, continuity, or welfare significance of the patient’s experiences.

A positive result is local to the target it tests.

3.4 Negative

A negative result is stronger than failure to obtain a positive result. It asserts that the assessment was applicable, adequately sensitive, properly executed, and yielded evidence within a validated rule-out region. This paper proposes the following negative license principle:

Negative License Principle: Absence of a consciousness indicator licenses a negative consciousness attribution only when the indicator was expected to be detectable if the target were present, the test was applicable to the system, auxiliary task demands were satisfied or controlled, data quality was adequate, and plausible alternative causes of non-detection were excluded.

When these conditions are not satisfied, “not detected” should not be translated into “absent.” This principle does not make negative attribution impossible. It makes its evidential burden visible.

3.5 Indeterminate

An indeterminate result occurs when the assessment genuinely bears on the question but does not discriminate enough to license either attribution or denial. Examples include scores in a validated gray zone, credible but conflicting indicators, and evidence that leaves multiple live explanatory models comparably supported.

Indeterminacy is therefore a result about the relation among target, evidence, and inferential standard. It is not a positive claim that the system occupies an ontologically intermediate state between consciousness and nonconsciousness. Consciousness may be binary, graded, multidimensional, or vague; the present framework is compatible with each possibility. It concerns what the assessment licenses.

3.6 Unlicensed

An unlicensed result occurs when the assessment cannot support the intended conclusion. Causes include poor signal quality, protocol failure, an unspecified target, an unstable system boundary, absent validation in the target population, or application outside the test’s substrate domain. “Unlicensed” is preferable to “indeterminate” in these cases because the test has not produced a valid intermediate consciousness result. It has failed to establish an interpretable relation to the target.

The distinction matters statistically and rhetorically. If unlicensed cases are placed in the negative class, specificity may appear high while false negatives are hidden. If they are pooled with valid indeterminate cases, the test’s limitations become harder to diagnose. Reporting them separately preserves the difference between a difficult case and an invalid inference.

4. The Indeterminacy Profile

A four-outcome label is still too coarse for cumulative science. Every indeterminate or unlicensed result should be accompanied by a structured cause profile. Five dimensions capture many recurrent problems.

4.1 Evidential scarcity

There may be too little reliable evidence to discriminate between presence and absence. The data can be valid yet weak: a score lies between thresholds, confidence intervals span both decision regions, or repeated trials yield high variance. The appropriate response is more or better-powered measurement, not theoretical reinterpretation.

Evidential scarcity must be distinguished from a negative result. Low signal can support absence only when the test's detection process is understood and the observed signal is lower than would be expected under consciousness. Otherwise, low signal is simply low information.

4.2 Surrogate discordance

Different measures intended to track the same target may conflict. Behavioral examination may indicate unresponsiveness while task-based neuroimaging indicates command following. Complexity measures may remain high while recurrent markers weaken. Two algorithms applied to the same recording may classify it differently.

Discordance can be scientifically productive because it exposes differences in task demand, temporal sensitivity, construct coverage, or confounding. It should be reported rather than averaged away unless the aggregation rule was justified and preregistered. The resolution path is measurement decomposition: determine whether the surrogates truly address the same target, identify their auxiliary demands, and use interventions or repeated measures to adjudicate the conflict.

4.3 Model indeterminacy

Evidence may be clear while its significance depends on a contested theory. A system might display recurrent processing but lack a global workspace; exhibit global availability but lack a higher-order representation; or possess several computational indicators while lacking biological features regarded as necessary by some theories. In such cases the indeterminacy profile should state which theoretical commitments produce which classifications.

This is not a reason to abandon theory-derived tests. It is a reason to avoid presenting a theory-conditional output as theory-independent. Adversarial tests and preregistered comparisons can reduce model indeterminacy by identifying observations on which rival theories make divergent predictions (Cogitate Consortium et al., 2025; Doerig et al., 2021).

4.4 Boundary and grain indeterminacy

Measures of integration, broadcast, complexity, and recurrence depend on which parts of a system are included and over what temporal window. A candidate system may contain nested or partially coupled organizations. Split brains, brain-body loops, organoid-computer interfaces, distributed robots, and modular artificial agents make this problem explicit, but ordinary neuroscience also selects spatial and temporal boundaries.

If reasonable boundary choices reverse the result, the attribution is boundary-sensitive. The appropriate response is not to select the favorable boundary after inspection. Investigators should preregister candidate boundaries, report sensitivity across them, and use perturbations to test causal autonomy. Until the attribution is stable or a principled individuation rule is supplied, the result remains indeterminate or unlicensed.

4.5 Transport indeterminacy

A test validated in one domain may not preserve its meaning in another. Physiological markers calibrated in adult humans may not transfer directly to infants, nonhuman animals, damaged brains, organoids, or artificial systems. Even within humans, etiology, medication, sensory impairment, and arousal can alter test sensitivity.

Transport indeterminacy is especially important in artificial consciousness. Indicator properties derived from neuroscientific theories offer a disciplined starting point (Butlin et al., 2023), but implementing an indicator is not equivalent to validating its evidential force in a new substrate. Structural or computational analogy can generate a research hypothesis without yet generating a licensed consciousness classification.

4.6 A reporting template

A compact report could take the following form:

Field	Required entry
Target	Exact form of consciousness being assessed
Candidate system	Boundary, grain, and time window
Test and model	Assessment battery and inferential assumptions
Validity envelope	Populations, substrates, and conditions covered
Outcome	Positive, negative, indeterminate, or unlicensed
Cause profile	Evidential, surrogate, model, boundary, and transport status
Robustness	Sensitivity to thresholds, preprocessing, and boundaries
Resolution path	Evidence or intervention most likely to change the result
Decision note	Practical action considered separately from scientific status

The resolution path is essential. A useful indeterminate result narrows the next question. “More research is needed” is not enough.

5. Worked Applications

5.1 Disorders of consciousness

Suppose a patient receives no diagnosis of command following at the bedside, but EEG data show a weak task-dependent response that does not cross a preregistered threshold. If the EEG recording is adequate and the task is appropriate, the result is indeterminate, not negative. Evidential scarcity is primary. If the patient has fluctuating arousal and the protocol was administered only once, the validity of a negative inference is further weakened; repeated standardized assessment is indicated (Giacino et al., 2018; Wannez et al., 2017).

Now suppose behavioral assessment repeatedly indicates unresponsive wakefulness while fMRI reveals reproducible command-following imagery. The result is not a simple disagreement to be averaged. The behavioral

test may be insensitive to covert command following because it requires motor output. For preserved command comprehension and volitional imagery, the evidence supports a positive attribution of at least some awareness-relevant capacity. Questions about ongoing phenomenal richness or continuity may remain indeterminate. The framework permits target-specific positive and indeterminate results to coexist.

Finally, suppose no task-based response is detected, but the patient cannot reliably hear instructions and sedating medication is present. The result is unlicensed for denial of consciousness. The protocol failed to establish the auxiliary conditions needed for a valid negative inference.

5.2 Nonhuman animals

Animal consciousness research often relies on clusters of behavioral, cognitive, anatomical, and physiological indicators rather than a decisive test. Theory-light approaches seek capacities that conscious perception plausibly facilitates while avoiding commitment to a single full theory (Birch, 2022). A multidimensional profile may also be more informative than ranking species on one consciousness scale (Birch et al., 2020).

Suppose an animal displays flexible learning, motivational tradeoffs, and protective behavior after injury, but the relevant neuroanatomy differs from that of mammals. The result may support sentience under a theory-light indicator framework while remaining transport-indeterminate for a mammal-calibrated neural marker. The proper report should preserve both facts. Failure to instantiate the mammalian mechanism is not automatically evidence of absence if functional alternatives remain plausible.

Conversely, a single behavior that can be generated by a simple reflexive mechanism may be insufficient to discriminate conscious from nonconscious processing. That is a valid indeterminate result if the behavior bears some relation to the target but lacks specificity. It should motivate tests designed to exclude the simpler explanation.

5.3 Neural organoids

Current organoids create an acute transport problem. Neural activity, oscillation, complexity, or perturbational responsiveness can be measured, but the interpretation of such measures was developed in systems with sensory histories, differentiated anatomy, vascular and regulatory support, and behavior. A human-like numerical value does not by itself establish equivalence of causal role.

If an organoid exhibits a proposed consciousness marker but lacks an established system boundary or the supporting organization required by the marker's theory, the consciousness result is unlicensed rather than positive. This does not prove absence. It states that the inferential bridge has not been validated. Research should focus on whether the marker is mechanistically homologous, whether relevant causal interactions exist, and whether the candidate system possesses the enabling conditions presupposed by the assessment.

5.4 Artificial systems

Artificial systems can produce fluent self-reports, implement recurrent computation, maintain global workspaces, or satisfy other theory-derived indicators. These features may be architecturally real even when their interpretation is disputed. The four-outcome framework blocks two opposite errors: dismissing every

artificial system because it lacks biological markers, and declaring consciousness whenever a computational analogue is identified.

An assessment of AI consciousness should report indicator evidence and transport status separately. If multiple theory-derived properties are credibly implemented but no cross-substrate calibration exists, the result may be model- and transport-indeterminate. If a test depends on biological processes absent from the system, its application is unlicensed rather than negative. If a theory explicitly predicts substrate-independent sufficiency and its specified conditions are met, the output can be positive relative to that theory while remaining model-indeterminate at the field level.

This conditional reporting is not evasive. It identifies precisely where disagreement lies: in implementation facts, substrate requirements, or the truth of the theory.

5.5 The N* model as an example

The N* model proposes that minimal phenomenal presence obtains exactly when a viable system instantiates integration or synergy, broadcast availability, and recurrent stability at the relevant grain (Stilwell, 2026). Its measurement policy already allows indeterminate classification when preregistered surrogates conflict. The present framework generalizes and sharpens that policy.

Under the four-outcome scheme, a system whose valid measures of integration disagree beyond tolerance is indeterminate. A system whose viability has not been assessed, whose graph boundary is selected retrospectively, or whose measures have not been validated for its substrate is unlicensed. A system is negative only when the full assessment is applicable and one or more proposed necessary conditions are robustly absent. This distinction protects the model from neither falsification nor counterexample; it specifies when an apparent counterexample is evidentially mature enough to bear that role.

6. Indeterminacy and Scientific Progress

6.1 Indeterminacy is informative when it is structured

A critic may object that science should reduce uncertainty, not institutionalize it. But an explicit indeterminate category can accelerate reduction of uncertainty by preventing heterogeneous failures from being pooled. The scientific value lies in decomposition. Evidential scarcity calls for more observations. Surrogate discordance calls for construct analysis and adjudicating interventions. Model indeterminacy calls for discriminating predictions. Boundary indeterminacy calls for causal individuation. Transport indeterminacy calls for bridge validation.

This converts a terminal statement - “we do not know” - into a research map. A mature field does not prove its rigor by forcing every measurement into a binary output. It proves rigor by matching the strength of its conclusions to the strength and domain of its evidence.

6.2 Indeterminate rates are properties of tests

The frequency of indeterminate and unlicensed outputs should be reported as part of test performance. A test that classifies only easy cases with high apparent accuracy may have limited usefulness if it returns no interpretable

result in the population where it is most needed. Conversely, a test with a wide indeterminate region may be appropriately calibrated when false classification is costly.

At minimum, validation studies should report:

- the proportions of positive, negative, indeterminate, and unlicensed outputs;
- the reasons for indeterminate and unlicensed classifications;
- performance estimates under transparent treatments of inconclusive cases;
- robustness across thresholds and repeated measurements;
- the population and substrate boundaries of the validity envelope.

Excluding indeterminate results from accuracy calculations can make a test appear more decisive than it is (Shinkins et al., 2013). Coding them as negative can conceal sensitivity failures. Coding unlicensed assessments as indeterminate can conceal non-applicability. All four outcomes matter.

6.3 Preregistration prevents opportunistic certainty

Thresholds, surrogate hierarchies, conflict rules, exclusion criteria, and system boundaries should be set before the critical result is examined. Otherwise, investigators can resolve ambiguity opportunistically by selecting a preferred measure, time window, preprocessing pipeline, or candidate boundary. Consciousness research is particularly vulnerable because many plausible measures exist and theories permit different post hoc interpretations (Yaron et al., 2022).

Preregistration cannot validate a poor test, but it can reveal whether certainty was earned by the protocol or manufactured after inspection.

7. From Scientific Status to Practical Action

Scientific classification and practical decision are related but distinct. A positive consciousness result normally strengthens reasons for protection. A negative result may weaken them. An indeterminate result, however, does not dictate either action or inaction. Decisions also depend on the possible magnitude of harm, reversibility, available alternatives, and the asymmetric costs of false attribution and false denial.

Let the scientific assessment produce an evidential state D , while a decision rule selects an action A by considering D together with a loss structure Λ :

$$A^* = \arg \min_A \mathbb{E}[\Lambda(A, C) \mid D].$$

This formula makes a simple point. Two institutions can accept the same indeterminate scientific result yet choose different actions because the stakes differ. Continuing a reversible, non-invasive measurement protocol is not equivalent to withdrawing life-sustaining treatment, exposing an animal to severe harm, scaling an organoid architecture, or deploying an artificial agent under conditions it describes as aversive. Precaution should be proportionate to both evidence and possible harm (Birch, 2024).

The separation also protects scientific language. Ethical precaution need not redescribe an indeterminate system as conscious. One can adopt safeguards because consciousness remains a credible possibility. Conversely,

scientific indeterminacy should not be invoked as a universal veto on research. The appropriate response can range from documentation and monitoring to redesign, independent review, or temporary restriction.

8. Objections and Replies

Objection 1: The framework merely renames uncertainty

The proposal would be trivial if it added only a third label. Its substantive content is the distinction between indeterminate and unlicensed outputs, the Negative License Principle, the validity envelope, the five-part cause profile, and the requirement that each unresolved result specify a resolution path. These constraints alter test validation, reporting, and inference.

Objection 2: Four outcomes weaken falsifiability

Any theory could be protected if every adverse result were called indeterminate. The remedy is preregistration. Applicability conditions, thresholds, conflict rules, and evidence required for positive and negative classification must be specified before outcomes are known. A result that falls in a preregistered negative region counts against a theory that predicted consciousness. Indeterminacy should not be available as a post hoc rescue category.

Objection 3: Consciousness is either present or absent

The framework does not deny bivalence at the level of reality. It distinguishes the state of the system from the state of warranted classification. A disease may be present or absent while a diagnostic result remains inconclusive. The same logical distinction applies to consciousness even if consciousness is metaphysically binary.

Objection 4: Probabilities already capture uncertainty

Probabilities can represent degrees of confidence but do not by themselves identify why confidence is limited or whether the test applies. A precise posterior produced by an invalid transport assumption remains unlicensed. Two cases can receive the same posterior for different reasons and require different next experiments. The cause profile supplements rather than competes with probabilistic analysis.

Objection 5: Model disagreement makes every hard case indeterminate

Not every disagreement is equally relevant. Some evidence is robust across theories, and some theories are better supported than others. The framework does not require unanimity. It requires that theory-conditional conclusions be labeled as such and that field-level conclusions reflect material, live disagreement. As theories gain or lose support, model indeterminacy can shrink.

Objection 6: The unlicensed category is too severe

“Unlicensed” does not mean scientifically worthless. An out-of-domain result may generate a valuable hypothesis or motivate calibration. It means only that the result does not yet warrant the intended consciousness attribution. This vocabulary protects exploratory evidence from being overinterpreted while preserving its role in discovery.

Objection 7: Precaution will bias the science toward positive attribution

Scientific outputs should not be altered to obtain preferred ethical consequences. That is why classification and action are separated. An indeterminate result remains indeterminate even when precautionary protection is justified. The framework is symmetrical at the scientific level and explicitly loss-sensitive at the decision level.

9. Recommendations

Researchers developing or applying consciousness tests should adopt the following minimum practices:

1. Specify the target form of consciousness rather than using “consciousness” without qualification.
2. Define the candidate system, spatial grain, temporal window, and relevant auxiliary capacities.
3. State the test’s validity envelope and identify all extrapolations beyond it.
4. Pre-register rule-in and rule-out thresholds, leaving an explicit indeterminate region where justified.
5. Separate valid indeterminate results from technically invalid or interpretively unlicensed assessments.
6. Report disagreements among surrogates before applying any composite score.
7. Attach an indeterminacy profile covering evidence, surrogates, models, boundaries, and transport.
8. State the observation or intervention most likely to resolve each identified source of indeterminacy.
9. Report the frequency and disposition of indeterminate and unlicensed cases in validation studies.
10. Keep scientific attribution separate from ethical or clinical action while making the decision rule and error costs explicit.

These practices are compatible with binary, graded, multidimensional, theory-heavy, and theory-light models of consciousness. They regulate the inference from evidence to attribution rather than dictating the correct theory of consciousness.

10. Conclusion

At the boundaries of consciousness science, forced binary classification creates an illusion of knowledge. A failed response becomes absence of experience; a conflicting biomarker is averaged away; an out-of-domain measure is treated as if its meaning traveled with its numerical value. These errors are avoidable.

Consciousness assessments should return four outcomes: positive, negative, indeterminate, and unlicensed. Negative attribution requires more than failure to detect a positive marker. Indeterminate results require a cause profile. Unlicensed results must be separated from valid gray-zone evidence. Each unresolved classification should identify a concrete resolution path.

The central claim is methodological but not modest in consequence: indeterminacy can be a successful scientific result. Properly structured, it identifies the exact limit of an inference, prevents false certainty, and tells investigators what kind of evidence would move the question forward. In a field without a universal gold standard and with increasingly consequential applications, knowing why a conclusion is not yet licensed is part of knowing what the evidence shows.

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